

Practical No. 10

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Batch: B2

Roll no:24

Aim:version control using Git& Github

What is difference between Git&Github

Ans : Git vs GitHub

Git:

Git is a Version Control System (VCS)

- Created by Linus Torvalds in 2005
- Works completely offline on your local machine
- Tracks every change made to your code
- Stores full project history locally
- Free and open-source
- Used via command line (terminal)
- Alternatives: Mercurial, SVN

GitHub:

GitHub is a cloud-based hosting platform for Git repos

- Founded in 2008, bought by Microsoft in 2018
- Requires internet connection
- Adds collaboration features: Pull Requests, Issues, Actions
- Stores your code remotely on servers
- Has free and paid plans
- Alternatives: GitLab, Bitbucket

2. Explain the difference between:

- **git add**

- **git commit**

Ans : git add:

- Moves files from Working Directory → Staging Area
- Tells Git which changes to include in next commit
- Does NOT save permanently
- Can stage one file, multiple files, or all files
- Reversible using `git restore --staged`

Commands:

`Git add file.txt` → one file

`Git add .` → all files

`Git add *.js` → all JS files

`Git add -p` → select parts interactively

git commit:

- Moves files from Staging Area → Local Repository
- Creates a permanent snapshot in history
- Each commit gets a unique SHA-1 hash ID
- Requires a commit message
- Records author, timestamp, and changes

Commands:

`Git commit -m "message"` → basic commit

`Git commit --amend` → edit last commit

`Git commit am "message"` → add + commit together

3. What is the staging area in Git?

Ans :

- Also called the Index
- It is a middle layer between working directory and repository

- Physically stored in .git/index file
- Lets you selectively choose what to commit
- You can modify 10 files but commit only 2
- Helps create clean, logical commits
- You can review staged changes before saving

Commands:

Git status → see staged/unstaged files

Git diff –staged → see staged changes

Git restore –staged file → unstage a file -

4 States of a file:

- Untracked → new file Git doesn't know about
- Modified → tracked file with new changes
- Staged → ready to be committed
- Committed → permanently saved in history

4. What happens if you modify a file after committing it?

Ans : The old committed version stays safe in history

- Modified file appears as “modified” in working directory
- Git detects change by comparing file hash with last commit
- Git status shows it under “Changes not staged for commit”
- You must add and commit again to save new version
- Each commit gets a new unique hash
- Old commit is still accessible anytime
- You can go back using git checkout <old-hash>
- Nothing in Git is truly lost after committing
- To fix last commit quickly: git commit –amend

Process after modifying:

Modify file → git add → git commit → new snapshot created

5. Difference between:

- **git push**

- **git pull**

Ans : git push:

- Uploads local commits → remote repository (GitHub)
- Shares your work with teammates
- Only pushes committed changes (not just staged)
- May be rejected if remote has newer commits

Commands:

Git push origin main → push to main branch

Git push origin branch-name → push specific branch

Git push -u origin main → set default upstream

Git push -force → force overwrite (dangerous)

Git pull:

- Downloads remote changes → local machine
- Keeps your local repo up to date
- Combines: git fetch + git merge
- Can cause merge conflicts if same lines were edited

Commands:

Git pull origin main → pull from main

Git pull -rebase → pull and rebase instead of merge

6. What is Version Control? Differentiate between Centralized v/s Distributed VCS.

Ans :

Version Control:

- System that records changes to files over time
- Tracks who, what, when, and why something changed
- Allows reverting to any previous version
- Enables parallel work without overwriting each other
- Essential for team software development

Centralized VCS (CVCS):

- One central server holds the entire repository
- Developers check out only one version (not full history)
- All operations need server/internet connection
- Examples: SVN, CVS, Perforce

Advantages:

- Simple to understand
- Admin has full control
- Good for binary/design files

Distributed VCS (DVCS):

- Every developer has full copy of entire repo including history
- Most operations happen locally (fast) • Remote server used only

for syncing/sharing

- Examples: Git, Mercurial

Advantages:

- Fully offline capability
- Every clone = complete backup
- Very fast operations
- Easy branching and merging
- No single point of failure

Theory -**Step by Step Procedure**

1. Installation of Git
2. Configure username & email
3. Create local repository
4. Create files
5. Add files to staging
6. Commit changes
7. Create GitHub repository
8. Link local repo to GitHub
9. Push files



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.gitignore

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